THIPPIREDDY MOHAN REDDY

AI / Computer Vision and Agentic AI | B.Tech IT (Business Informatics), IIIT Allahabad | CGPA 8.22

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PROFESSIONAL SUMMARY

Second-year B.Tech student at IIIT Allahabad and research member of the CIR Robotics Research Group, working across computer vision, robotics and agentic AI. Experience covers the full ML stack: video point tracking for surgical vision, detection, segmentation and depth pipelines, probabilistic supervision of deep models, and multi-agent LLM systems with guardrails. Comfortable benchmarking under strict latency budgets, designing evaluation protocols, and taking prototypes to deployment. 3rd globally at STIRC 2026 (MICCAI EndoVis); 2nd nationally at Strategy Wiz (MNNIT Allahabad). Seeking research and engineering internships in AI, computer vision and agentic systems.

TECHNICAL SKILLS

- **Languages:** Python, C, C++
- **Machine Learning and Deep Learning:** PyTorch, TensorFlow/Keras, scikit-learn, LightGBM, XGBoost, CNNs, Transformers (self-attention, RoPE), fine-tuning, knowledge distillation, pseudo-label and self-supervised training, loss design, cross-validation, hyperparameter tuning, feature engineering, Bayesian inference, Kalman filtering, uncertainty estimation
- **Computer Vision:** OpenCV, YOLOv8, object detection, instance segmentation, point tracking (LiteTracker, CoTracker3, Track-On2, DenseTrack2D, MFT), optical flow (RAFT), monocular depth (Depth-Anything-V2), stereo vision, DINOv2, video understanding, medical imaging
- **Agentic AI and LLMs:** multi-agent orchestration, LLM APIs (Gemini, Groq), prompt engineering, guardrails and response validation, LLM-as-judge, Hugging Face Transformers
- **Tools:** Git, Docker, Linux, CUDA, Jupyter, Colab, Kaggle, NumPy, Pandas, SciPy, Matplotlib, ONNX, TensorRT, GitHub Actions, FFmpeg

RESEARCH EXPERIENCE

CIR Robotics Research Group, IIIT Allahabad (Prof. G.C. Nandi) | 2026 - Present

- Researching "Generative Affordance and Predictive World-Model Guidance for Dexterous Articulated Manipulation": using generated affordance predictions and world models to guide manipulation of articulated objects.
- Contribute to the group's video point-tracking research (SurgQueST), including a pipeline audit that found a core implementation gap, independently validated by a labmate.

PROJECTS

STIR Challenge 2026: Surgical Tissue Tracking (MICCAI EndoVis) | Team IIIT Allahabad | Python, PyTorch | 3rd globally

- Co-authored the 2D-track methodology report for a Track-On2 based submission; benchmarked 10+ trackers (DenseTrack2D, CoTracker3, MFT, RoMa) under a 200 ms/frame streaming budget.
- Built and ablated correction layers for online tracking (Kalman-predicted handoff, confidence-gated re-anchoring, keyframe correction); Kalman handoff matched a costly re-anchoring gain at +0.4% latency.
- Developed a length-stratified 60-sequence evaluation set and a two-eye stereo pipeline.

HBIPT: Probabilistic Supervisory Layer for Point Tracking (on LiteTracker) | Python, PyTorch

- Designed a five-layer supervisor (working memory, evidence space, recursive Bayesian belief filter over 8 hypothesis families, gated actions) that monitors a tracker at runtime and can abstain; with actions disabled it is bit-identical to the stock tracker on 60/60 sequences at +1.1% latency.
- Showed that standard nearest-point scoring hides identity failures: strict index matching drops AJ by ~0.30 across four trackers and at least 8% of predictions collide on one landmark.
- Evaluated each intervention with paired bootstrap CIs and oracle analysis; ported MFT's backcheck as an online module (0.80 precision vs 0.70 break-even).

CogAIt: Socratic Multi-Agent Learning Platform | React, Convex, Node.js, Gemini, Groq, Cloud Run

- Built an 8-agent orchestration (tutor, analyst, content, mastery, ingestion, cohort, recommendation, judge) that scales help across four levels without revealing answers.
- Added guardrails: a think-first gate, a rate-limited proxy, and a judge agent that audits responses for answer leakage; validated with 10 runtime-invariant suites and a live stress test.

AI-Powered Road Damage Detection and Severity Triage | Python, PyTorch, YOLOv8, Depth-Anything-V2

- Built a detection, segmentation and severity-rating pipeline (ASTM D6433-based) that outputs structured JSON for a municipal dashboard.
- Merged and deduplicated 4 public datasets into 1,986 images with a leakage-free split; raised recall from 0.68 to 0.86 and mAP@50:95 from 0.48 to 0.70; 5-fold CV mAP@50 0.73; 2.3% background false-positive rate.

Autonomous Content Engine | Python, GitHub Actions, Groq (LLaMA 3.1), FFmpeg, YouTube/Instagram/Facebook APIs

- Built a fully automated pipeline that generates scripts, renders 1080x1920 video and uploads to three platforms on a schedule, with validation, retries and failure isolation.
- Added a weekly analytics loop: an LLM analyzes performance data and feeds new content ideas back into the pipeline.

SiteBook: Site Attendance and Wage Tracking App | React Native (Expo), Firebase, Cloud Firestore

- Built a mobile app for attendance, overtime, advances and monthly PDF/Excel pay reports, with per-user security rules, duplicate-resistant writes, PIN lock and optional Google Drive backup.

EDUCATION

Indian Institute of Information Technology, Allahabad | 2025 - Expected 2029

B.Tech, Information Technology (Business Informatics) | CGPA 8.22

Coursework: Operating Systems, Computer Organization and Architecture, Discrete Mathematics, Machine Learning, Deep Learning

CERTIFICATIONS

- Deep Learning Specialization, Andrew Ng, DeepLearning.AI. Verify: deeplearning.ai/certificates/0a9c8e89-d559-431c-9a19-ce1ee0566b3e

ACHIEVEMENTS

- 3rd globally (team), STIR Challenge 2026, MICCAI EndoVis
- 2nd nationally, Strategy Wiz, MNNIT Allahabad, 2025
- JEE percentile: 99.363

EXTRACURRICULAR ACTIVITIES

Member, E-Cell Sponsorship Team, IIIT Allahabad | 2025 - Present